

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2016-17

Paper No. GLB-301: Petrology and Economic Geology

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem III/169

*Instructions: Attempt **five** questions in all, selecting **two** questions from Section A and **two** questions from Section B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Describe the magmatic differentiation and assimilation processes. [14]

Question 2. Write short notes on **any four** of the following: $[4 \times 3.5 = 14]$

- (a) Cross-bedding
- (b) Classification of Sandstone
- (c) Dolerite
- (d) Gneiss
- (e) Metamorphic textures

Question 3. What do you understand by clastic texture? Discuss in detail the observable properties based on which the texture of a clastic sedimentary rock is described. [14]

Question 4. Briefly describe the types of metamorphism. Discuss the metamorphic Facies. [14]

SECTION — B

Question 5. Write in detail the different classifications of ore deposits. [14]

Question 6. What is coal? Discuss the stages of coal formation and the Indian distribution of bituminous coal. [14]

Question 7. Write the chemical compositions, important physical properties, and uses of **any four** of the following economic minerals: $[4 \times 3.5 = 14]$

- (a) Magnetite
- (b) Talc
- (c) Chromite

- (d) Kyanite
- (e) Graphite
- (f) Galena

SECTION — C (Compulsory)

Question 8. Comment on the following in two or three sentences each: [7 × 2 = 14]

- (i) Write the four divisions of the hydrothermal process.
- (ii) Difference between ore and gangue with examples.
- (iii) Draw neat sketches of folded and faulted oil traps.
- (iv) Define metasomatism. Write the essential minerals of syenite rock.
- (v) Difference between block lava (aa) and ropy lava (pahoehoe).
- (vi) Difference between oomicrite and oosparite.
- (vii) Intraformational conglomerate.